Delft University of Technology

GEO5017

Assignment 2

Objects Classification from Airborne LiDAR Data

Group Members

Stijn Brouwers – 5398681
Merijn Brons – 5411815
Maarten Hulsman – 5379989

Instructor

Dr. L. (Liangliang) Nan

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Introduction

The objective of this assignment is to classify pre-segmented airborne LiDAR point clouds into five urban-object categories: building, car, fence, pole, and tree. To solve this task, we designed geometric and distributional features for each object, selected the four most discriminative ones, and compared two supervised classifiers introduced in the course: Support Vector Machine (SVM) and Random Forest (RF).
The assignment required us to define an initial feature set, apply a quantitative feature-selection strategy, optimize both classifiers through hyperparameter tuning, and evaluate their performance using learning curves and confusion matrices. This report describes the feature design process, the selected features, the final classifier settings, and the comparison between SVM and RF.

Methodology

Initial feature design

For each segmented point cloud $P = \{p_i\}_{i=1}^{N}$, $p_i = (x_i, y_i, z_i) \in \mathbb{R}^3$, a 36-dimensional feature vector was computed using $\varepsilon = 10^{-5}$ for numerical stability. The basic size descriptors are

$$\text{width} = \max_i x_i, \qquad \text{depth} = \max_i y_i, \qquad \text{height} = \max_i z_i.$$

Let p_{root} and p_{top} be the points with minimum and maximum z-coordinate, respectively. Root density is defined as the fraction of points within planar radius 0.2 of p_{root}:

$$\text{root_density} = \frac{1}{N} \left| \{p_i \in P : \|(x_i, y_i) - (x_{\text{root}}, y_{\text{root}})\|_2 \leq 0.2\} \right|.$$

From the 2D convex hull of $\{(x_i, y_i)\}_{i=1}^{N}$, we computed

$$\text{hull_area} = A_{\text{hull}}, \qquad \text{shape_index} = \frac{A_{\text{hull}}}{P_{\text{hull}}},$$

where A_{hull} and P_{hull} are the hull area and perimeter. Around p_{top}, the

$$k_{\text{top}} = \max\big(\lfloor 0.005N \rfloor, 100\big)$$

nearest neighbors were extracted, and the eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ of their covariance matrix (obtained via PCA) were used to compute geometric features commonly used in point cloud classification Thomas et al., 2018

$$\text{sum_eigen} = \lambda_1 + \lambda_2 + \lambda_3, \quad \text{omnivariance} = (\lambda_1\lambda_2\lambda_3)^{1/3},$$

$$\text{eigenentropy} = -\sum_{m=1}^{3} \lambda_m \ln(\lambda_m + \varepsilon), \quad \text{linearity} = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \varepsilon}, \quad \text{planarity} = \frac{\lambda_2 - \lambda_3}{\lambda_1 + \varepsilon},$$

$$\text{sphericity} = \frac{\lambda_3}{\lambda_1 + \varepsilon}, \qquad \text{curvature} = \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3 + \varepsilon}.$$

These eigenvalue-based features describe the local geometric structure of point clouds and are widely used in LiDAR classification tasks Thomas et al., 2018. In addition, $\text{n_points} = N$.
Three elongation ratios were computed as

$$\text{elong_xy} = \frac{\max(\text{width}, \text{depth})}{\min(\text{width}, \text{depth}) + \varepsilon}, \quad \text{elong_yz} = \frac{\max(\text{depth}, \text{height})}{\min(\text{depth}, \text{height}) + \varepsilon}, \quad \text{elong_xz} = \frac{\max(\text{width}, \text{height})}{\min(\text{width}, \text{height}) + \varepsilon},$$

and point density in the bounding box as

$$\text{volume} = \frac{N}{\text{width} \cdot \text{depth} \cdot \text{height} + \varepsilon}.$$

Finally, after translating the cloud by its coordinate-wise minima, normalized 6-bin histograms were computed along the x-, y-, and z-axes, yielding 18 profile features `prof_x_0` to `prof_x_5`, `prof_y_0` to `prof_y_5`, and `prof_z_0` to `prof_z_5`. Altogether, this gives $3 + 1 + 2 + 7 + 1 + 3 + 1 + 18 = 36$ features. These descriptors were chosen to capture differences in size, footprint, local top geometry, compactness, and vertical point distribution between buildings, cars, fences, poles, and trees.

Feature selection

To identify the four most informative features, we used a Fisher-style class-separability measure based on between-class and within-class scatter. For a feature subset S, the score is defined as:

$$J(S) = \operatorname{tr}\left((S_W + \epsilon I)^{-1} S_B\right) \tag{1}$$

where S_W is the within-class scatter matrix, S_B is the between-class scatter matrix, and ϵI is a small regularization term for numerical stability.
First, each feature was scored individually using its Fisher score. The ten highest-ranked individual features were `prof_z_5` (1.38494), `root_density` (1.34645), `height` (1.17092), `prof_z_3` (1.13776), `curvature` (1.07268), `sphericity` (0.94377), `prof_z_2` (0.93733), `linearity` (0.90465), `planarity` (0.64566), and `shape_index` (0.63818). After that, greedy forward selection was used to construct a subset of exactly four features by iteratively adding the feature that produced the largest improvement in the subset score. The final selected feature subset was:

$$\{\texttt{prof_z_5},\ \texttt{prof_z_3},\ \texttt{prof_z_2},\ \texttt{root_density}\} \tag{2}$$

The final subset achieved a Fisher score of 6.82630. This result suggests that vertical density information is highly discriminative for this task. In particular, the selected z-profile features describe how the points are distributed over the height of the object, while root density captures the concentration of points near the ground. The ranking also shows that height, curvature, sphericity, and linearity are individually useful, but that the best four-feature combination is dominated by vertical-profile information rather than by simple size descriptors alone.

Classification models

After feature selection, both classifiers were trained on the selected four-dimensional feature space.

- **Support Vector Machine (SVM):** Several kernels and hyperparameter combinations were tested using grid search. The best-performing model used an RBF kernel with $C = 100$ and $\gamma =$ `auto`. This suggests that the class boundaries in the selected feature space are non-linear and benefit from a flexible kernel.

- **Random Forest (RF):** Different values for the number of trees, maximum tree depth, and minimum samples required for a split were evaluated. The best configuration used `n_estimators` $= 50$, `max_depth` $= 10$, and `min_samples_split` $= 5$. This suggests that a moderately sized ensemble with controlled tree depth gives a better balance between flexibility and generalization than either very shallow or fully unconstrained trees.

A train-test split of 60:40 was used for the final reported models. Learning curves were generated by repeatedly training the models with training-set ratios between 0.1 and 0.9. The resulting accuracies were used to assess model generalization behavior.

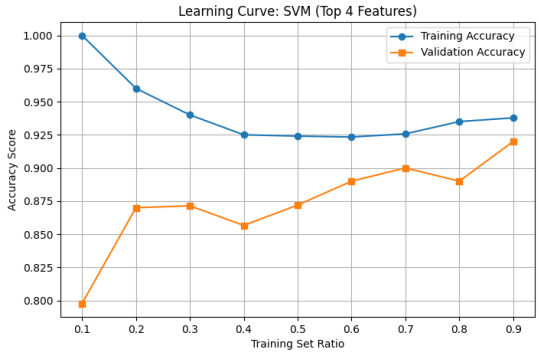
Results

The selected four features were all related to vertical structure and near-ground point density. This indicates that the internal height distribution of the point cloud is more informative than simple bounding-box dimensions alone.
Table 1 summarizes the most important quantitative outputs of the feature-selection and classification stages.

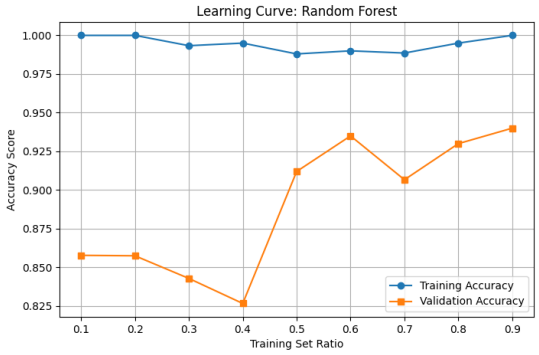
Item	**Result**
Top 3 individual Fisher features	`prof_z_5`, `root_density`, `height`
Selected 4-feature subset	`prof_z_5`, `prof_z_3`, `prof_z_2`, `root_density`
Subset Fisher score	6.82632
Best SVM model	$kernel = "linear"$, $C = 100$, $\gamma =$ `scale`
Best RF model	`n_estimators`=200, `max_depth`=20, `min_samples_split`=5
Optimization-stage accuracy (SVM)	0.890
Optimization-stage accuracy (RF)	0.930
Final reported overall accuracy (SVM)	0.94
Final reported overall accuracy (RF)	0.9850

Table 1: Summary of feature-selection and classifier results.

The SVM and RF classifiers both produced usable results, but Random Forest clearly performed better overall. During hyperparameter optimization, the SVM obtained an accuracy of 0.82 on the test set, while the optimized RF obtained 0.91. This already indicated that RF could exploit the selected feature subset more effectively. The learning curves are shown in Figures 1a and 1b. The confusion matrices of the final classifiers are shown in Figures 2a and 2b.
The SVM learning curve shows that the training accuracy remains around 0.90–0.96, while validation accuracy fluctuates around 0.77–0.85. This indicates a moderate generalization gap and suggests that SVM is not able to separate all classes consistently in the selected feature space.
The Random Forest learning curve shows near-perfect training accuracy and a much stronger validation performance, reaching approximately 0.94 for the larger training-set ratios. This shows that RF can model the structure of the data more effectively than SVM.



(a) Learning curve of the final SVM classifier.

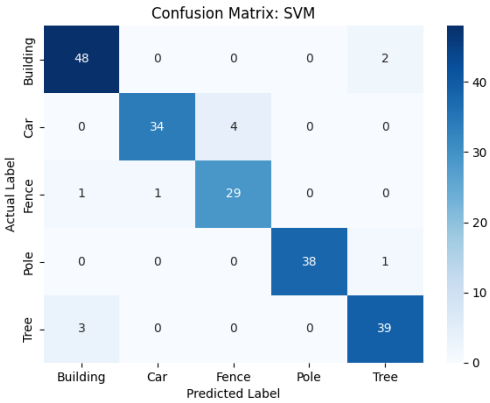


(b) Learning curve of the final Random Forest classifier.

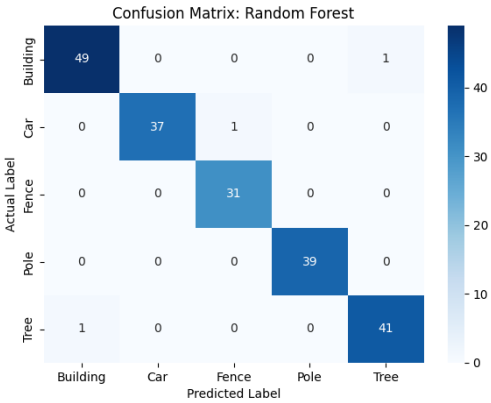
Figure 1: Learning curves of the two final classifiers.

The confusion matrix of SVM shows that most errors occur between *building* and *tree*. In particular, many buildings are classified as trees. There are also smaller confusions between *car* and *fence*, and a few pole

predictions are confused with buildings. This suggests that SVM struggles with class boundaries that are not easily separable in the selected feature space.
The Random Forest confusion matrix is much cleaner. Buildings and poles are classified almost perfectly, while only a few errors remain between cars and fences and between trees and buildings. This confirms that RF is better able to model the non-linear relationships in the data.
For the final comparison between both classifiers, overall accuracy (OA) was selected as the main performance metric. This choice is appropriate because the dataset is balanced across the five classes and because OA was consistently used throughout the experiments and learning-curve analysis. To complement OA, confusion matrices were used for class-specific error analysis. The available evaluation outputs show a clear overall advantage for Random Forest: during hyperparameter optimization, RF achieved a higher test accuracy (0.91) than SVM (0.82). In the final classification reports, RF also achieved a higher overall accuracy (0.9800) than SVM (0.8600). Based on the confusion matrices, the learning curves, and the overall model behavior, Random Forest is the preferred classifier for this task.



(a) Confusion matrix of the final SVM classifier.



(b) Confusion matrix of the final Random Forest classifier.

Figure 2: Confusion matrices of the two final classifiers.

Conclusion

This assignment showed that airborne LiDAR object classification can be performed effectively using a compact set of carefully selected geometric descriptors. The feature-selection procedure revealed that vertical density profiles and root density are highly informative for distinguishing between the five object classes.
Between the two tested classifiers, Random Forest produced the strongest results. It achieved better validation performance, cleaner confusion matrices, and more robust overall behavior than SVM. The quantitative outputs from both the optimization stage and the final evaluation also consistently favored RF over SVM. The main remaining classification difficulties were between trees and buildings, and to a lesser extent between cars and fences.
Possible improvements for future work include adding more shape descriptors, testing repeated cross-validation instead of a single split, evaluating feature normalization strategies more explicitly, and designing additional features that better separate trees from buildings.

Division of the tasks

The task division within the group was organized as follows:
Stijn Brouwers

- Feature engineering and preprocessing
- Support with methodology structuring and report integration

Merijn Brons

- Feature selection and classifier tuning
- Generating evaluation outputs, including learning curves and confusion matrices

Maarten Hulsman

- Writing the Introduction, Results, and Conclusion sections
- Integrating and checking the final report

Experiment

Our code can be found here (M. Brons, S. Brouwers, M. Hulsman, 2026).

References

M. Brons, S. Brouwers, M. Hulsman. (2026). *Geo5017*. https://github.com/mgahulsman/geo5017/tree/main/Assignment2

Thomas, H., Goulette, F., Deschaud, J.-E., Marcotegui, B., & LeGall, Y. (2018). Semantic classification of 3d point clouds with multiscale spherical neighborhoods. *2018 International Conference on 3D Vision (3DV)*, 390–398.